18.3: Noncoding RNAs play multiple roles in controlling gene expression

Genome sequencing has revealed that protein-coding DNA accounts for only 1.5% of the human genome and a similarly small percentage of the genomes of many other multicellular eukaryotes. A very small fraction of the non-protein-coding DNA consists of genes for RNAs such as ribosomal RNA and transfer RNA. Until recently, scientists assumed that most of the remaining DNA was not transcribed, thinking that since it didn't specify proteins or the few known types of RNA, such DNA didn't contain meaningful genetic information—in fact, it was called "junk DNA." However, some genomic studies have cast doubt on this description. For example, a massive study showed that roughly 75% of the human genome is transcribed at some point in any given cell. Introns account for only a fraction of this transcribed RNA, most of which is untranslated. The verdict is still out on how much of the transcribed RNA is functional, but at least some of the genome is transcribed into non-protein-coding RNAs (also called *noncoding RNAs*, or *ncRNAs*), including a variety of small RNAs. Researchers are uncovering more evidence of the biological roles of these ncRNAs every day.

These discoveries have revealed a large and diverse population of RNA molecules in the cell that play crucial roles in regulating gene expression and have gone largely unnoticed until recently. The longstanding view that mRNAs are the most important RNAs because they code for proteins needs revision. This represents a major shift in thinking by biologists, one that you are witnessing as students entering this field of study.

Effects on mRNAs by MicroRNAs and Small Interfering RNAs

Regulation by both small and large ncRNAs occurs at several points in the pathway of gene expression, including mRNA translation and chromatin modification. We'll examine two types of small ncRNAs, the importance of which was acknowledged when their discovery was the focus of the 2006 Nobel Prize in Physiology or Medicine, which was awarded for work completed only eight years earlier.

Since 1993, a number of research studies have uncovered **microRNAs** **(miRNAs)**—small, single-stranded RNA molecules capable of binding to complementary sequences in mRNA molecules. A longer RNA precursor is processed by cellular enzymes into an miRNA, a single-stranded RNA of about 22 nucleotides that forms a complex with one or more proteins **(Figure 18.15)**. The miRNA allows the complex to bind to any mRNA molecule with at least seven or eight nucleotides of complementary sequence. The miRNA-protein complex then degrades the target mRNA or, less often, simply blocks its translation. There are

approximately 1,500 genes for miRNAs in the human genome, and biologists estimate that expression of at least one-half of all human genes may be regulated by miRNAs, a remarkable figure given that the existence of miRNAs was unknown until the early 1990s.

Figure 18.15

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    Regulation of gene expression by microRNAs (miRNAs).
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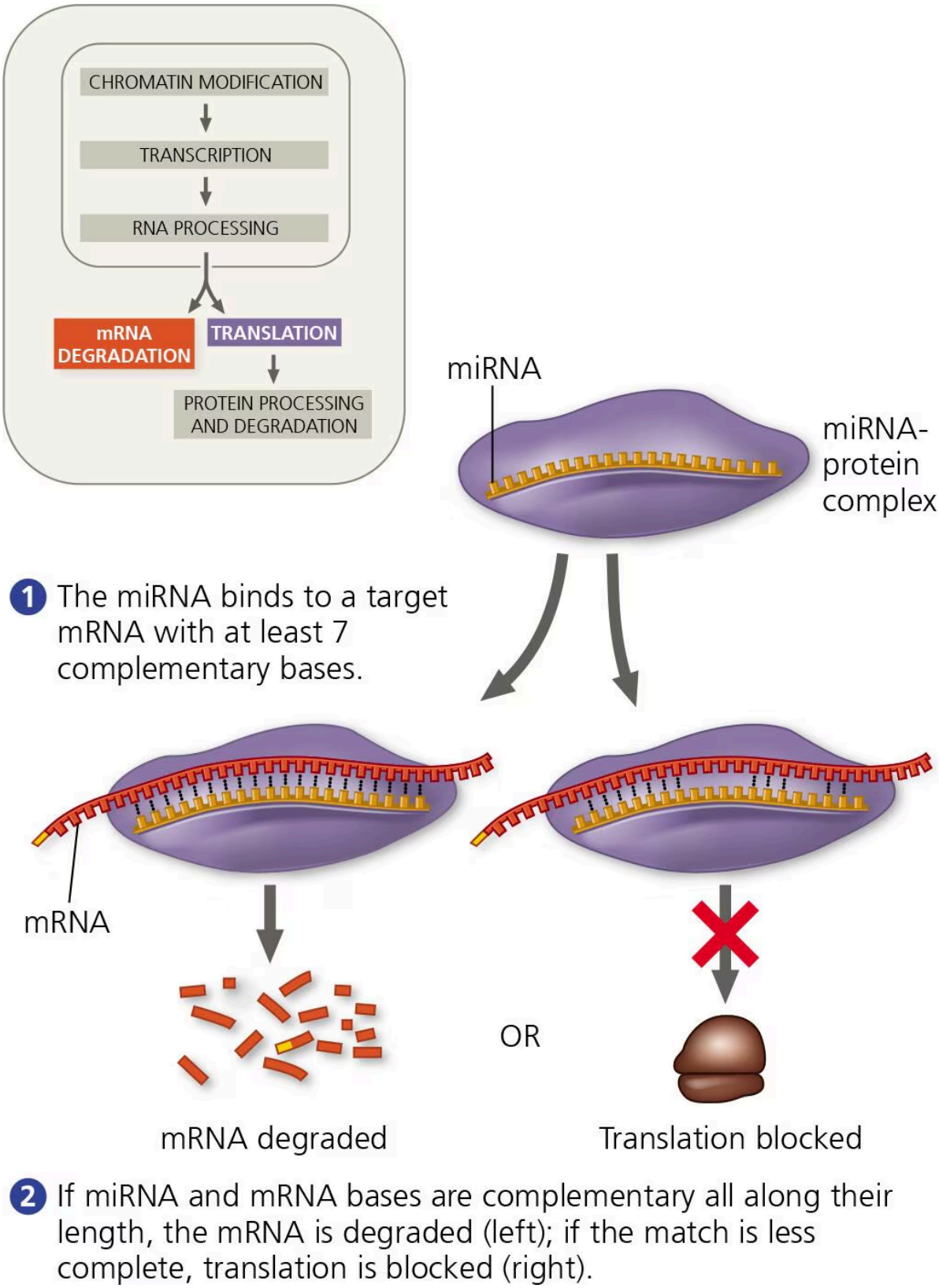

An miRNA of about 22 nucleotides, formed by enzymatic processing of an RNA precursor, associates with one or more proteins in a complex that can degrade or block translation of target mRNAs.

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  – Overview: Stages in gene expression subject to regulation detailed in a flowchart: W
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- 2. If m i R N A and m R N A bases are complementary all along their length, the m R N A is degraded. Degraded m R N A shown in pieces. If the match is less complete, translation is blocked. Crossed out arrow shows blocked path to translation in the ribosome.

Another class of small noncoding RNAs, similar in size and function to miRNAs, is called **small interfering RNAs (siRNAs)**. Both miRNAs and siRNAs can associate with the same proteins, producing similar results. In fact, if siRNA precursor RNA molecules are injected into a cell, the cell's machinery can process them into siRNAs that turn off expression of genes with related sequences, similarly to how miRNAs function. The distinction between miRNAs and siRNAs is based on subtle differences in the structure of their precursors, which in both cases are RNA molecules that are mostly double-stranded. The blocking of gene expression by siRNAs, referred to as **RNA interference (RNAi)**, is used in the laboratory as a means of disabling specific genes to investigate their function.

EVOLUTION How did the RNAi pathway evolve? As you will learn in Concept 19.2, some viruses have double-stranded RNA genomes. Given that the cellular RNAi pathway can process double-stranded RNAs into homing devices that lead to destruction of related RNAs, some scientists think that this pathway may have evolved as a natural defense against infection by such viruses. However, the fact that RNAi can also affect the expression of nonviral cellular genes may reflect a different evolutionary origin for the RNAi pathway.

While this section has focused on ncRNAs in eukaryotes, small ncRNAs are also used by bacteria as a defense system, called the CRISPR-Cas9 system, against viruses that infect them (see Concept 17.5). The use of ncRNAs thus evolved long ago, but we don't yet know how bacterial ncRNAs are related to those of eukaryotes.

Chromatin Remodeling and Effects on Transcription by ncRNAs

In addition to regulating mRNAs, small noncoding RNAs can cause remodeling of chromatin structure. In the S phase of the cell cycle, for example, the centromeric regions of DNA must be loosened for chromosomal replication and then recondensed into heterochromatin in preparation for mitosis. In some yeasts, siRNAs produced by the yeast cells from the centromeric DNA are required to re-form the heterochromatin at the centromeres. Exactly how the process starts is still debated, but biologists agree on the general idea: The siRNA system in yeast interacts with other, larger noncoding RNAs and with chromatin-modifying enzymes to condense the centromere chromatin into heterochromatin. In most mammalian cells, siRNAs have not been found, and the mechanism for centromere DNA condensation is not yet understood. However, it may also turn out to involve small noncoding RNAs.

A recently discovered class of small ncRNAs called *piwi-interacting RNAs*, or *piRNAs*, also induce formation of heterochromatin, blocking expression of some parasitic DNA elements in the genome known as transposons. (Transposons are discussed in Concept 21.4.) Usually 24–31 nucleotides in length, piRNAs are processed from a longer, single-stranded RNA precursor. They play an indispensable role in the germ cells of many animal species, where they appear to help reestablish appropriate methylation patterns in the genome during gamete formation.

Interview

Interview with Haifan Lin: Discovering piwi-interacting RNAs and connections to testicular cancer

Researchers have also found a relatively large number of **long noncoding RNAs (lncRNAs)**, ranging from 200 to hundreds of thousands of nucleotides in length, that are expressed at significant levels in specific cell types at particular times. The functional significance of these lncRNAs has been debated, but in 2017 a large international research consortium published an atlas of almost 28,000 such RNAs; their analysis supported the idea that almost 20,000 were functional and that some were associated with specific diseases. One lncRNA, long known to be functional, is responsible for X chromosome inactivation, which prevents expression of genes located on one of the X chromosomes in most female mammals (see Figure 15.8). In this case, lncRNAs—transcripts of the *XIST* gene located on the chromosome to be inactivated—bind back to and coat that chromosome. This binding leads to condensation of the entire chromosome into heterochromatin.

The examples just described involve chromatin remodeling in large regions of the chromosome. Because chromatin structure affects transcription and thus gene expression, RNA-based regulation of chromatin structure is sure to play an important role in gene regulation. Additionally, some experimental evidence supports the idea of an alternate role for lncRNAs in which they can act as a scaffold, bringing DNA, proteins, and other RNAs together into complexes. These associations may act either to condense chromatin or, in some cases, to help bring the enhancer of a gene together with mediator proteins and the gene's promoter, activating gene expression in a more direct fashion.

Given the extensive functions of noncoding RNAs, it is not surprising that many of the ncRNAs characterized thus far play important roles in embryonic development—the topic we turn to in the next section. Embryonic development is perhaps the ultimate example of precisely regulated gene expression.